

Data Analytics (DA)

Module -3

MySQL Assignment 2 - Querying Data

Assignment Task Overview

Problem Statement:

As a Data Analyst, you will continue working with the **Employee Database** created in the previous assignment. The database already contains the required tables, relationships, and constraints.

Use this DML insert code, [DML Insertion- Employee Data](#), to populate the tables

The focus of this assignment is on **data insertion, querying, filtering, aggregation, and result interpretation** while ensuring consistency with the existing database design.

Dataset Description:

- **Table: *Departments***

Attributes:

- *department_id – Primary Key*
- *department_name*

- **Table: *Location***

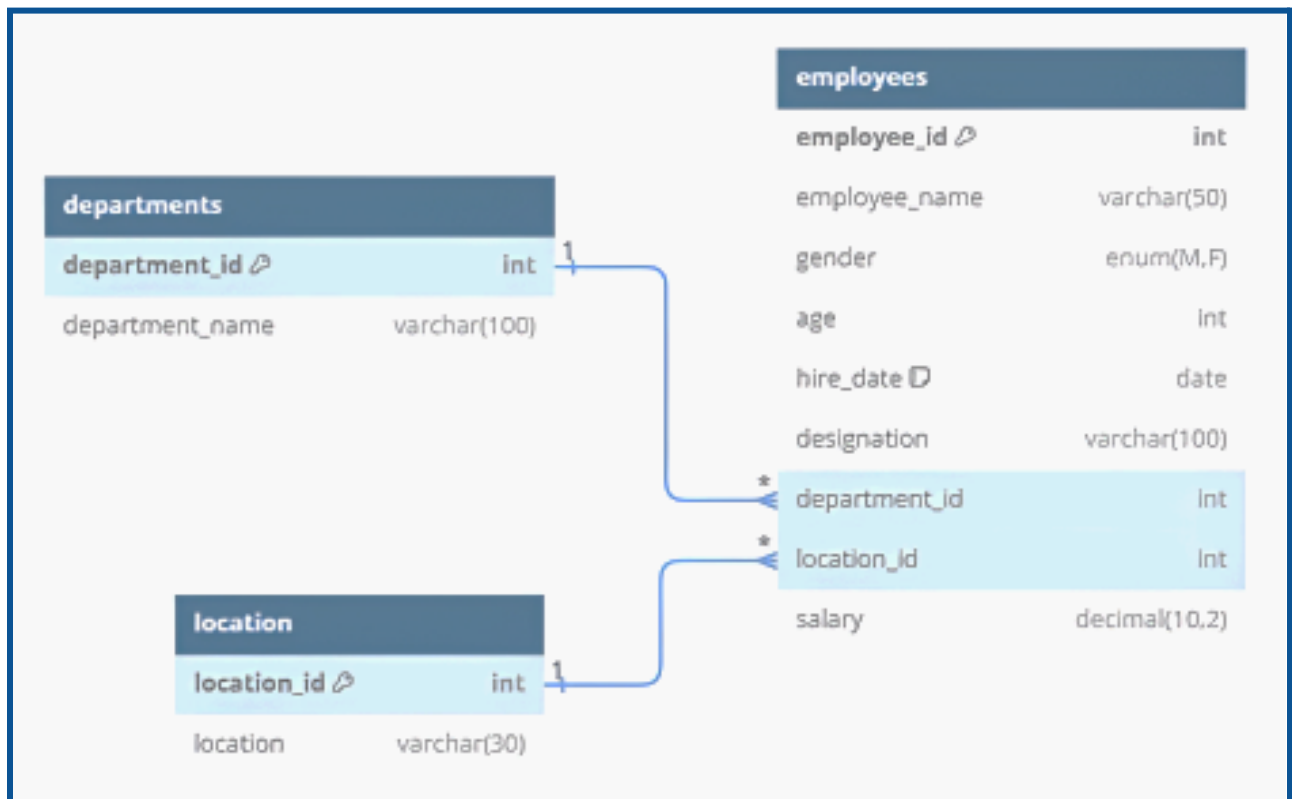
Attributes:

- *location_id – Primary Key*
- *location_name*

- **Table: *Employees***

Attributes:

- *employee_id – Primary Key*
- *Employee_name*
- *Gender*
- *Age*
- *Hire_date*
- *Designation*
- *Salary*
- *department_id – Foreign Key referencing department(department_id)*
- *location_id—Foreign Key referencing location(location_id)*



Tasks

Clause & Operators:

1. DISTINCT VALUES:

- A query to retrieve distinct salaries from the Employees table.

2. ALIAS (AS):

- Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

3. WHERE CLAUSE & OPERATORS:

- Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.
- Find the employee whose designation is missing and fill it with "Data Scientist".

Sorting and Grouping Data:

1. ORDER BY:

- Find employees sorted by department ID in ascending order and salary in descending order.

2. LIMIT:

- Display the first 5 employees hired in the year 2018.

3. AGGREGATE FUNCTIONS:

- Calculate the sum of all salaries in the Finance department.
- Find the minimum age among all employees.

4. GROUP BY:

- List the maximum salary for each location.
- Calculate the average salary for each designation containing the word 'Analyst'.

5. HAVING:

- Find departments with less than 3 employees.
- Find locations with female employees whose average age is below 30.

Joins:

1. INNER JOIN:

- List employee names, their designations, and department names where employees are assigned to a department.

2. LEFT JOIN:

- List all departments along with the total number of employees in each department, including departments with no employees.

3. RIGHT JOIN:

- Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

4. CROSS JOIN

- Show all possible combinations of departments and locations.

5. SELF JOIN:

- Show pairs of employees working in the same department, excluding self-pairs.

Windows function

- Write a window function query to rank employees by salary using rank().
- Write a window function query to rank employees by salary within each department using DENSE_RANK()
- Write a window function query, Running total salary by department

Deliverables

Submit a project folder containing:

1. **MySQL Script (.sql)** that includes:
 - Queries demonstrating Clauses & Operators
 - i. Sorting data using ORDER BY
 - ii. Grouping data using GROUP BY
 - iii. Filtering grouped data using HAVING
 - Join Queries
 - i. INNER JOIN
 - ii. LEFT JOIN
 - iii. RIGHT JOIN
 - iv. CROSS JOIN
 - v. SELF JOIN
 - Windows functions
2. **PDF Document** that includes:
 - Execution screenshot of the queried output

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Proficiency
DISTINCT & ALIAS	Data Retrieval	DISTINCT, Column Aliasing (AS)	Beginner
WHERE Clause & Operators	Data Filtering	WHERE, Comparison Operators, NULL Handling	Beginner
Sorting Data	Result Set Ordering	ORDER BY (ASC, DESC)	Beginner
Row Limiting	Result Control	LIMIT, Date-based Filtering	Beginner
Aggregate Functions	Data Aggregation	SUM, MIN, AVG	Beginner
GROUP BY	Data Grouping	GROUP BY with Aggregates	Intermediate
HAVING Clause	Group Filtering	HAVING with Aggregate Conditions	Intermediate
INNER JOIN	Table Relationships	Matching Records Across Tables	Beginner
LEFT JOIN	Outer Joins	Handling Missing Related Records	Intermediate
RIGHT JOIN	Outer Joins	Location-based Data Retrieval	Intermediate
CROSS JOIN	Cartesian Product	Generating all combinations of departments and locations	Intermediate
SELF JOIN	Table Self-Referencing	Pairing employees within the same department	Intermediate
Windows function	Query Structuring	Ranking, partitioning, and running calculations using window functions	Intermediate
Analytical Thinking	Business Logic	Translating Business Questions into SQL	Intermediate

Rubric & Mark Allocation (Total 10 Marks)

Skill / Component	Marks	Criteria for Full Marks	Partial Credit Description	Marks Range
DISTINCT & ALIAS Usage, WHERE Clause & Operators	2	Correct use of DISTINCT and column aliases Accurate filtering using salary, date, and NULL handling	Syntax correct, but alias/DISTINCT misuse Logical condition partially incorrect	1-1.5
Aggregate Functions GROUP BY & HAVING ORDER BY & LIMIT	4	Correct use of SUM, MIN, AVG Correct grouping and filtering on aggregates Proper sorting and row limiting applied	Aggregate logic is partially incorrect GROUP BY correct, HAVING incorrect The sorting order or the year filter is incorrect	1 – 3
JOIN Operations	2	Correct INNER, LEFT, RIGHT, CROSS & SELF joins with valid outputs	The join condition is partially incorrect	1-1.5
Windows function	2	Window functions are implemented correctly	Window function syntax is correct but logic is partially incorrect(e.g., incorrect partitioning, ordering)	1-1.5

DA Skill Taxonomy Structure

- **Data Retrieval:** SELECT, DISTINCT, column aliases
- **Data Filtering:** WHERE, operators, NULL handling, date filters
- **Sorting & Limiting Results:** ORDER BY, LIMIT
- **Aggregation & Grouping:** SUM, MIN, AVG with GROUP BY
- **Group-Level Filtering:** HAVING with business conditions
- **Table Relationships:** INNER, LEFT, RIGHT JOIN, CROSS JOIN, SELF JOIN

- **Windows functions-** Rank(),DenseRank(),PARTITION BY ,ORDER BY
- **Business Interpretation:** Convert business requirements into SQL logic
- **SQL Query Proficiency:** Clean, optimised, industry-standard MySQL queries

Evaluation Guidelines

- **Open and review the submitted .sql script** to ensure:
 - Tables are correctly created and populated.
 - All data exploration Using Joins and five analytical queries are present and executable.
- **Check the correctness of joins, grouping, and calculations.**
- **For each section,** match student work to the specific skill/sub-skill area in the taxonomy.
- **Assign full or partial marks** according to the rubric.
 - Full marks: accurate syntax, logic, and clean formatting.
 - Partial marks: minor issues, incomplete queries, or inconsistent naming.
- **Comment** on strengths and improvement points for each section.
- **Maintain objectivity**—assess the correctness, clarity, and relevance of SQL queries.

Achievement Criteria & Passing Rules

- At least partial (non-zero) marks must be obtained in every assignment section.
- Students must fully achieve at least two major skills: SQL Querying & Data Analysis Skills, Relational Database handling and real-world business questions into SQL logic
- Feedback and a **detailed scorecard** must be provided for every evaluated submission.